# Chapter 5
# A Systematic Review of AI's Effect on Students' Mental Health and Well-Being

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## ABSTRACT

*This systematic review synthesises recent empirical evidence on the impact of artificial intelligence (AI) technologies on the mental health and well-being of higher education students. Drawing on 40 peer-reviewed studies published between January 2023 and August 2025, the review examines interventions and tools that employ AI, including chatbots (n = 10), generative models (n = 6), wearables (n = 7), emotion-recognition systems (n = 9), and adaptive learning platforms (n = 8). Following PRISMA 2020 guidelines and using the SPIDER framework, studies were selected based on their empirical focus on AI-mediated educational or psychological support. Findings indicate that AI-enabled interventions (particularly those grounded*

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*in cognitive-behavioural therapy) consistently reduce symptoms of depression and anxiety, and improve problem-solving, self-efficacy and perceived social support. AI-based measurement tools also demonstrate promising reliability.*

## INTRODUCTION

Higher education institutions (HEIs) worldwide are witnessing an unprecedented convergence of two powerful forces: a worsening student mental health crisis and the rapid proliferation of artificial intelligence (AI) technologies (Charles & Alshamsi, 2025). Meta-analyses of the World Health Organisation (WHO) World Mental Health Surveys International College Student project reveal that more than one-third of first-year students screened positive for at least one DSM-IV disorder (35.3 % lifetime prevalence; 31.4 % 12-month prevalence) (Auerbach et al., 2018). The persistence and early age-of-onset of disorders such as major depression and anxiety during the formative years of tertiary study not only undermine academic achievement but also jeopardise young adults' long-term wellbeing and socio-economic prospects.

AI technologies offer novel mechanisms to deliver, augment, and monitor mental health support at scale. For clarity, this review adopts a broad definition of AI, encompassing machine learning and natural language systems deployed in education, including conversational agents and chatbots, large language models, adaptive or predictive analytics, and affective computing (Charles, 2025). We conceptualise student mental health and wellbeing following the WHO's holistic framework, encompassing the absence of psychopathology (e.g., depression, anxiety, stress, loneliness) and the presence of positive attributes such as psychological resilience, life satisfaction and social support.

### Purpose of the Study

Multiple researchers have explored the promise of AI-enabled mental-health interventions since the launch of ChatGPT in November 2022 (Azzam & Charles, 2025). For example, a proof-of-concept study demonstrated that university students who engaged with the 'Atena chatbot' over four weeks (receiving psychoeducation and coping exercises) experienced significant reductions in anxiety and stress, especially those with higher baseline anxiety (Gabrielli et al., 2021). Another example can be found in a single-blind randomised controlled trial conducted by He et al. (2022), wherein students assigned to 'XiaoE', an AI-driven cognitive-behavioural therapy chatbot, reported significantly lower depression scores at one-week and one-month follow-ups compared with peers using e-books or general chatbots.

Similarly, an unblinded trial found that a chatbot-delivered self-help intervention achieved greater reductions in depression and anxiety than bibliotherapy, and users reported a stronger therapeutic alliance with the chatbot (Liu et al., 2022). Beyond intervention studies, evidence from a global survey of students using the 'Replika' chatbot revealed that, although users were lonelier than typical students, many felt they received high social support, and 3% credited the chatbot with preventing suicidal ideation (Maples et al., 2024). Furthermore, according to Gu et al. (2025), AI may also enhance 'measurement', because in their cross-sectional study, they found that ChatGPT-generated assessments of loneliness and online social support correlated closely with validated scales (intraclass correlations of 0.81 and 0.95, respectively).

These studies suggest that AI applications have the potential to alleviate psychological distress (Charles & Papadaki, 2025), foster engagement and augment assessment. Yet the evidence base is fragmented across diverse technologies, populations, outcomes and methodological quality. No comprehensive synthesis has yet examined the scope, effectiveness and mechanisms of AI-driven interventions or predictive tools for students' mental health. Given the urgent need to address student wellbeing and the rapid adoption of generative AI, a systematic review is warranted to map existing evidence, evaluate efficacy, identify risks and guide policy and practice.

## Research Questions

Accordingly, this systematic review is guided by the following research questions:

- RQ1: What AI applications have been employed within higher-education settings to influence or monitor students' mental health and wellbeing?
- RQ2: What are the reported effects of these AI applications on psychological outcomes such as depression, anxiety, stress, loneliness, resilience and life satisfaction?
- RQ3: Which factors moderate or mediate the impact of AI on student mental health?
- RQ4: Where do gaps and methodological limitations remain in this literature?

By addressing these questions, the review aims to provide an evidence-informed foundation for educators, counsellors and policymakers seeking to harness AI responsibly to support student wellbeing and to highlight areas where caution or further development is required.

## LITERATURE REVIEW

AI-enabled mental health tools in higher education are predominantly underpinned by cognitive behavioural therapy (CBT). CBT conceptualises psychological distress as a product of maladaptive cognitions and behaviours; interventions aim to modify these through psychoeducation, cognitive restructuring and behavioural activation. Early chatbot trials have operationalised CBT protocols within conversational agents (He et al., 2022; Liu et al., 2022); the findings of these studies suggest that AI chatbots can establish a working alliance and deliver CBT content effectively, aligning with longstanding evidence on the importance of therapeutic engagement in psychological change. Beyond CBT, social support frameworks help interpret the emergence of AI companions (Maples et al., 2024). Such outcomes are consistent with social-support theory, which posits that perceived companionship and emotional support can buffer stress and reduce risk. Measurement frameworks are also evolving (Gu et al., 2025), and the high reliability of AI-generated scores suggests the potential of AI systems as psychometric tools, an area grounded in classical test theory and item-response theory.

Globally, university students face significant mental-health challenges (Abusneneh & Charles, 2023). Major depressive disorder and generalised anxiety disorder are among the most common conditions, often beginning in early adolescence and persisting into tertiary education (Auerbach et al., 2018). Traditional campus counselling services struggle to meet this demand, particularly in low- and middle-income countries and during crises such as the COVID-19 pandemic. At the same time, students are increasingly engaging with AI-enabled technologies for learning, entertainment, and personal assistance (Charles & Hill, 2025). These trends have spurred interest in AI-mediated mental-health interventions as scalable, on-demand supports.

Empirical evidence on the impact of AI on student mental health is growing, but remains limited. For example, the XiaoE trial, mentioned earlier, demonstrated that AI-mediated CBT can outperform static resources, with participants reporting both symptom reduction and a positive user experience (He et al., 2022). Also, in the unblinded trial by Liu et al. (2022), students using the chatbot experienced greater improvements in depression and anxiety compared with a bibliotherapy control, and their therapeutic alliance ratings were higher (Liu et al., 2022). Open and quasi-experimental studies provide complementary insights (Maples et al., 2024). These observations raise questions about the mechanisms through which AI companions confer benefits—whether through CBT-style psychoeducation, unconditional positive regard, or the mere presence of an empathetic interlocutor. Moreover, the study on ChatGPT-generated loneliness and social-support scores highlights an emerging line of research that leverages large language models as assessment tools (Gu et al., 2025). The strong agreement with established instruments suggests that AI could streamline the monitoring of student well-being in educational settings.

While these studies have shown positive outcomes, it is essential to interpret them through theoretical lenses that explain the cognitive and affective mechanisms underlying them. In this chapter, Cognitive Load Theory (CLT) is used as a sensitising framework for understanding how AI tools may alleviate—or exacerbate—stress in higher-education contexts. Cognitive load refers to the amount of information working memory can process at a time and is often described in terms of intrinsic, germane, and extraneous components. On one hand, AI tools such as adaptive learning systems and intelligent tutoring can reduce peripheral cognitive demands through personalised feedback and scaffolding, thereby supporting engagement and well-being. On the other hand, poorly designed AI interfaces, prompt-heavy workflows, or excessive notifications may increase extraneous load, contributing to confusion, fatigue, demotivation, anxiety, and mental strain (Azazz, 2025; Song, 2025).

The technostress perspective complements cognitive load theory by explaining psychological strain associated with technology use and overuse. University students can experience technostress as anxiety, digital fatigue, and reduced academic performance, particularly when digital systems demand constant connectivity or rapid adaptation (Galvin et al., 2020; Klimova & Pikhart, 2025; Lițan, 2025). AI-mediated educational tools, while beneficial, can inadvertently contribute to technostress when students are overwhelmed by notifications, algorithmic demands, or persistent online engagement. In short, CLT helps us understand why students may feel overwhelmed when learning tasks are complex and information-rich, whereas technostress highlights how socio-technical pressures can further amplify mental strain. CBT-oriented tools (including AI-supported chatbots) can be conceptualised as one coping route that supports emotion regulation and mental resilience within digitally intensive learning environments.

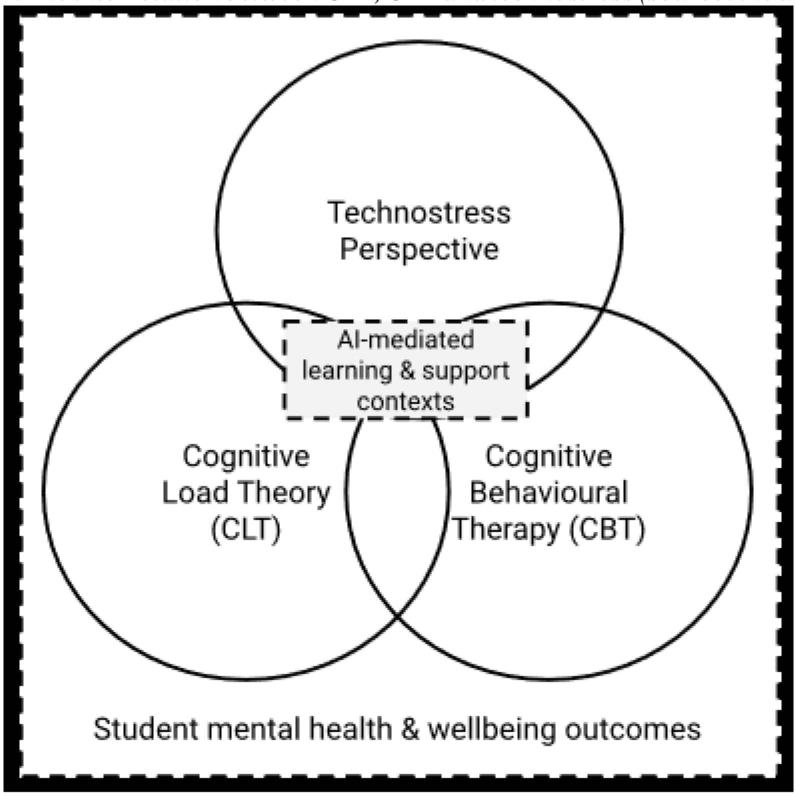
Integrating these frameworks with empirical findings from CBT-based chatbots highlights a comprehensive picture (*see* Figure 1): AI interventions can suppress depressive and anxiety symptoms and foster perceived social support, yet the broader AI ecosystem in higher education carries the risk of cognitive overload and technostress. Despite these promising findings, the literature exhibits several limitations. Sample sizes are small (often fewer than 150 participants), follow-ups are short, and participant demographics are not always diverse. Comparators vary widely (from bibliotherapy to minimal interventions), making it challenging to synthesise effect sizes. Few studies assess long-term outcomes, adverse effects (e.g., over-reliance, technostress) or integration with existing counselling services. Moreover, the majority of interventions rely on CBT, leaving other therapeutic paradigms (e.g., acceptance and commitment therapy, interpersonal psychotherapy) unexplored. The heterogeneity of AI technologies—from rule-based chatbots to generative models—also complicates comparisons. Finally, the ethical implications of deploying AI in student mental health care, including data privacy, bias, and the potential erosion of human connection, are acknowledged but not systematically investigated in the studies mentioned above.

## AI Use as Cyber Behaviour and Digital Coping

Because this volume foregrounds cyber behaviours, it is important to frame students' engagement with AI not solely as the use of a tool but as a form of digitally mediated behaviour, encompassing help-seeking, self-disclosure, emotion regulation, and social interaction through conversational agents. This behavioural lens also underscores that AI systems are embedded within datafied learning environments: students generate behavioural trace data (e.g., prompts, interaction logs, wearable signals, smartphone-derived markers) that can be analysed through learning analytics, affective analytics, and related forms of digital phenotyping to infer well-being states and risk profiles (Aalbers et al., 2023; Prinsloo et al., 2024). Accordingly, AI can simultaneously function as (a) an intervention that offers coping resources (e.g., CBT scripts, mindfulness prompts, companionship) and (b) a socio-technical stressor that reshapes attention, autonomy, and social connectedness (Selim et al., 2024; Maples et al., 2024).

However, the benefits and risks of AI-mediated cyber behaviours are unlikely to be evenly distributed across student populations. Students with disabilities or special educational needs may face accessibility barriers or distinctive well-being pressures in digitally intensive learning contexts (Abusneneh & Charles, 2023), and minoritised or marginalised students may be disproportionately exposed to algorithmic bias, surveillance harms, or reduced access to culturally responsive support. Conceptual work on vulnerable student digital well-being in AI-powered educational decision support systems emphasises the need for fiduciary responsibility, inclusive governance, and equity-oriented design (Prinsloo et al., 2024). Related scholarship also notes that techno-positive attitudes toward AI can coexist with concerns about how AI shapes quality of life and participation for people with disabilities and other groups (Lillywhite & Wolbring, 2024).

*Figure 1. The interrelation between CLT, CBT and technostress (source: the authors)*




In summary, the existing literature suggests that AI-driven chatbots grounded in CBT can help reduce depressive and anxiety symptoms among university students and may also foster perceptions of social support (He et al., 2022). AI systems also show promise as psychometric tools, with high reliability compared with traditional questionnaires (Gu et al., 2025). However, the field is nascent, and current evidence is limited by small sample sizes, short durations, and restricted theoretical diversity. These gaps underscore the need for a systematic synthesis to identify effective AI applications, understand mechanisms of action and inform ethical deployment in higher education.

In summary, the existing literature suggests that AI-driven chatbots grounded in CBT can help reduce depressive and anxiety symptoms among university students and may also foster perceptions of social support (He et al., 2022). AI systems also show promise as psychometric tools, with high reliability compared with traditional questionnaires (Gu et al., 2025). However, the field is nascent, and current evidence is limited by small sample sizes, short durations, and restricted theoretical diversity. These gaps underscore the need for a systematic synthesis to identify effective

AI applications, understand mechanisms of action and inform ethical deployment in higher education. Building on the theoretical framing above—including CLT, technostress, CBT, and a cyber behaviour perspective—the next section outlines the systematic review methods used to identify, screen, and synthesise evidence on AI applications that influence or monitor student mental health and well-being in higher education.

## METHODOLOGY

### Procedures for Systematic Review

This systematic review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guideline (Page et al., 2021). PRISMA 2020 emphasises that systematic reviews must provide a transparent, complete and accurate account of why the review was done, how studies were identified and selected, and what results were found (Liberati et al., 2009). It is underpinned by a 27-item checklist and updated to reflect advances in evidence identification, synthesis methods, and risk-of-bias assessment (Moher et al., 2015).

Of the 307 screened records, 64 articles were retrieved for full-text assessment. Following the eligibility review, 24 studies were excluded, primarily because they were out of scope (e.g., not focused on student mental health or not involving AI-based interventions or measurements). A final set of 40 studies met the inclusion criteria and were retained for synthesis. The screening and inclusion process is illustrated in the PRISMA 2020 flow diagram (*see* Figure 2). Inter-rater agreement at the abstract screening stage was approximately 92%, and disagreements were resolved through discussion and consultation with a third reviewer when necessary.

Eligibility criteria were formulated using the SPIDER framework (Amir-Behghadami, 2021). Studies were included if they: (1) involved Samples of higher-education students (undergraduate or postgraduate); (2) examined a Phenomenon involving AI used in teaching, learning, assessment or student support (e.g., generative AI, chatbots, adaptive or predictive systems, affective computing); (3) the Interest is to investigate the AI applications and their effects on mental health in higher education (4) employed empirical Designs (randomised trials, quasi-experimental, observational, qualitative or mixed-methods); (5) reported Evaluation outcomes related to mental health or wellbeing (e.g., depression, anxiety, stress, loneliness, psychological wellbeing, resilience); and (6) were of any Research type (quantitative, qualitative or mixed). We excluded studies conducted in non-higher education settings, studies involving non-AI technologies, purely theoretical or editorial pieces, commentaries, and protocols; studies published before January 1, 2023; and studies not in English.

*Table 1. Database search parameters applied.*

| Academic Database | Parameters Applied | Example Search Strategy (Boolean/Keywords) | Records Retrieved |
|---|---|---|---|
| **APA PsycInfo** | Filters: Peer-reviewed, Academic Journals, English; Research articles only; Education domain | (“artificial intelligence” OR AI OR “large language model” OR LLM) AND (“mental health” OR anxiety OR depression OR wellbeing OR stress) AND (“university student” OR “higher education” OR undergraduate OR postgraduate) | 4 |
| **ERIC** | Peer-reviewed; Education-only filter applied | (“artificial intelligence” OR AI) AND (“mental health” OR “well-being”) AND (“higher education” OR university OR student) | 31 |
| **PubMed** | Human studies; English; Published from 2023 | (“artificial intelligence” OR AI) AND (“depression” OR “anxiety” OR “stress” OR “mental health”) AND (“university students” OR “higher education”) | 166 |
| **Scopus** | Peer-reviewed journals; English; Subject areas: Social Sciences, Psychology, Health | TITLE-ABS-KEY(“artificial intelligence” OR AI OR LLM) AND TITLE-ABS-KEY(“mental health” OR stress OR depression OR well-being) AND TITLE-ABS-KEY(“higher education” OR university OR student) | 108 |
| | | | Total: 309 |

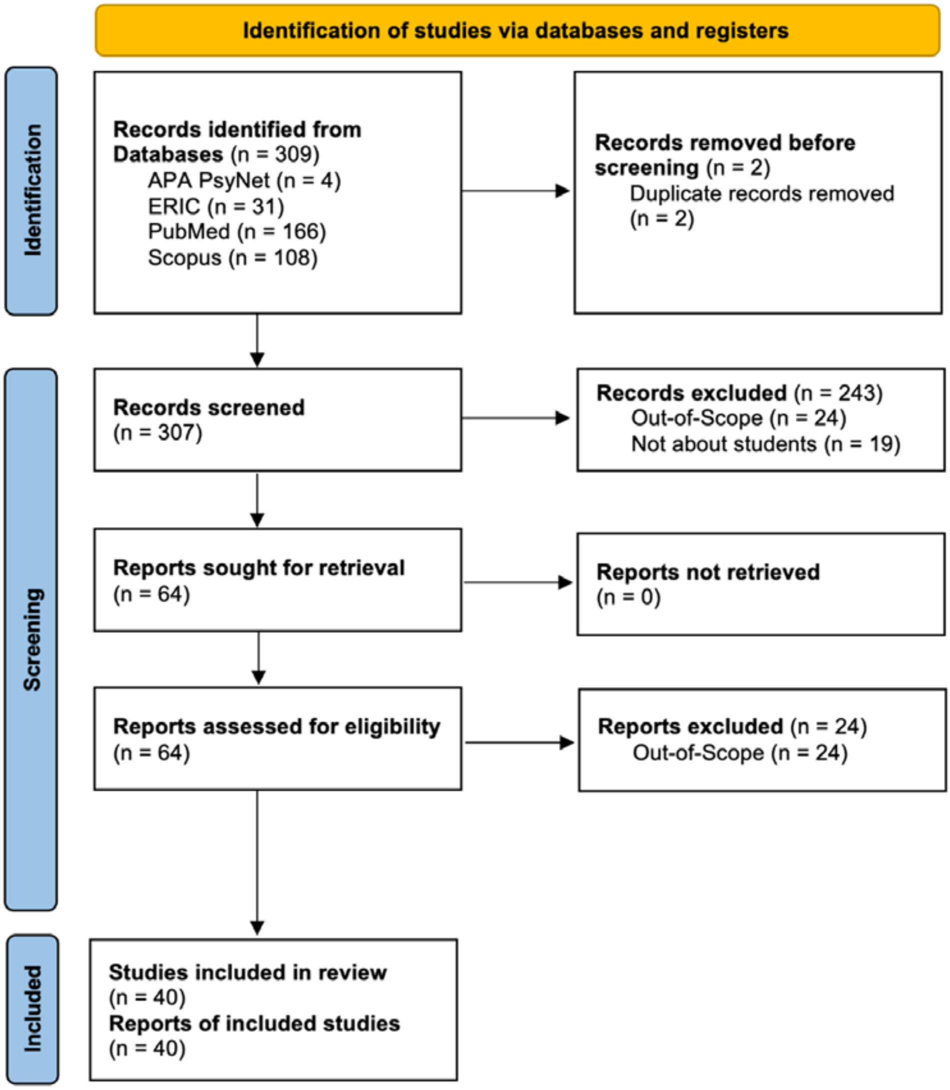
The search strategy was developed iteratively and tailored for each database (*see* Table 1). Four databases—Scopus, ERIC, APA PsycINFO, and PubMed— were searched to capture literature published between January 1, 2023, and August 9, 2025. Following the PRISMA guidance on transparent reporting of search methods (Page et al., 2021), two search variants were executed per database: a broad “Recall+” query and a narrower “Precision+” query. Queries combined controlled vocabulary and free-text keywords for three concept clusters, ensuring the Boolean operator ‘AND’ was used to combine them: (1) AI technologies (“*artificial intelligence*”, “*AI*”, “*large language model*”, “*LLM*”); (2) higher-education populations (“*student*”, “*undergraduate*”, “*postgraduate*”, “*higher education*”, “*university*”); and (3) mental-health outcomes (“*mental health*”, “*wellbeing*”, “*well-being*”, *“anxiety”*, “*technostress*”). Searches were limited to peer-reviewed journal articles in English.

## Procedures for Document Analysis

Duplicate references were identified and removed using a combination of exact DOI matching and fuzzy title matching. In total, 309 records were retrieved; deduplication removed two duplicates, leaving 307 unique records for screening (*see* Figure 2). A two-stage screening process was undertaken in accordance with

PRISMA 2020 recommendations (Moher et al., 2015). Two reviewers independently screened titles and abstracts against the eligibility criteria. The inter-rater agreement at the abstract screening stage was approximately 92%; disagreements were resolved through discussion. Any record judged as "include" or "uncertain" by either reviewer proceeded to full-text screening. Full-text articles were independently assessed by the same reviewers (i.e. the two lead authors of this chapter); disagreements were resolved through discussion and, when necessary, consultation with a third reviewer (i.e. the third listed author of this chapter). Reasons for exclusion at the full-text stage were recorded (e.g., incorrect population, non-AI, incorrect outcomes, non-empirical, outside date range). Data were synthesised narratively, and studies were grouped by AI application (e.g., CBT-based chatbots, generative AI companions, predictive analytics) and by mental-health outcomes. Qualitative findings were synthesised using thematic analysis to identify recurring themes relating to user experience, perceived benefits, harms and contextual factors. The integration of quantitative and qualitative findings followed a convergent, segregated approach.

*Figure 2. PRISMA flow diagram, informed by Page et al. (2021).*




## RESULTS

Table 2 provides an aggregated snapshot of the evidence base, while Appendix A provides a detailed, study-by-study overview of the included literature. The thematic synthesis generated five overarching themes: (1) AI-based interventions and monitoring for mental health and well-being; (2) patterns of AI use and adoption; (3) reported psychological outcomes and benefits; (4) reported risks and unintended consequences; and (5) ethical, policy, and institutional considerations. Each theme is discussed below.

*Table 2. Aggregated overview of AI application categories and outcome focus in the included evidence base*

| AI application category | Typical implementation in HE | Primary outcome focus | Example studies |
|---|---|---|---|
| Intervention chatbots (CBT / counselling support) | Automated dialogue, CBT-informed prompts, self-help exercises; delivered via apps/ web | Depression, anxiety, distress, self-efficacy, problem-solving | Wrightson-Hester et al., 2023; Kang & Hong, 2025; Reyes-Portillo et al., 2025 |
| Generative-AI companions & prospection chatbots | LLM-mediated companionship, reflection, goal-setting and future-oriented prompts | Loneliness/social support, agency, engagement, well-being perceptions | Dechant et al., 2025; Maples et al., 2024; Zhang et al., 2025 |
| Mindfulness & well-being platforms | AI-supported mindfulness and nature-based interventions; physical activity/PE technologies | Psychological well-being, enjoyment, emotional regulation | Wang et al., 2024; Xu, 2025 |
| AI-enhanced learning tools & SEL | Adaptive feedback/scaffolding, AI writing support, AI-enhanced social-emotional learning | Cognitive load, academic engagement, digital well-being | Song, 2025; Shi et al., 2025; Zong & Yang, 2025; Wu, 2024 |
| Wearables, physiological sensing & social robots | Wearable sensing for stress detection; embodied/robotic support for relaxation | Stress indicators, self-regulation cues, personalised recommendations | Abd-Alrazaq et al., 2024; Ahmed et al., 2025; Klęczek et al., 2024 |
| Measurement & analytics tools | AI-adapted scales and predictive modelling using behavioural/physiological data | Digital life balance, depression/ideation risk, well-being profiling | Erdemir & Atik, 2025; Meda et al., 2023; Aalbers et al., 2023; Gu et al., 2025 |
| Ethics, governance & institutional implementation | Policy, fiduciary duty, participatory design, institutional guidance and workforce development | Privacy, bias, accountability, equitable access, implementation feasibility | Delello et al., 2025; Prinsloo et al., 2024; Skorburg et al., 2024; Connell, 2025 |

## Theme 1 – AI-Based Mental-Health and Well-Being Interventions

Under Theme 1, the evidence clustered into two complementary subthemes: (a) intervention-based AI, where systems deliver psychoeducational or therapeutic content to support coping and self-regulation; and (b) assessment and monitoring

AI, where systems infer or track well-being indicators using behavioural and/or physiological data.

## Theme 1(a) – Intervention-Based AI

The most extensively investigated intervention modality involved conversational agents. Studies of CBT-informed chatbots (e.g., MYLO and Dr CareSam) suggest that structured dialogue flows can support self-management of depression and anxiety, alongside improvements in problem-solving and self-efficacy, particularly when content is bounded and aligned with psychoeducational principles (Wrightson-Hester et al., 2023; Kang & Hong, 2025). More recent generative AI chatbots (e.g., Wayhaven and Future Me) broadened affordances toward reflective conversations, prospection, and goal setting; however, evaluations also note risks of formulaic responses, limited emotional nuance, and declining engagement over time (Reyes-Portillo et al., 2025; Dechant et al., 2025). Student narratives further indicate that AI-mediated help-seeking can function as a cyber behaviour—supporting anonymous disclosure and emotional rehearsal—but should be positioned as complementary to, rather than substitutive of, human counselling and peer support (Maples et al., 2024; Selim et al., 2024).

Beyond chatbots, intervention-based AI was embedded within broader well-being and learning ecosystems. An AI-powered web platform combining nature-based multimedia with mindfulness exercises reduced anxiety and improved sleep quality, illustrating how AI can support self-regulation through structured content rather than open-ended dialogue (Wang et al., 2024). AI-enabled physical education technologies similarly enhanced enjoyment and psychological well-being (Xu, 2025). In the learning domain, AI-enhanced tools and AI-supported writing systems were associated with improved engagement, self-regulated learning, and well-being outcomes, with cognitive load acting as a key design-relevant mechanism (Song, 2025; Shi et al., 2025; Wu, 2024). Finally, AI-enhanced social-emotional learning frameworks show promise for combining pedagogical scaffolding with affective support and preventive well-being aims (Zong & Yang, 2025).

## Theme 1(b) – Assessment and Monitoring AI

A second cluster of studies used AI to detect, predict, or track well-being states. Wearable- and sensor-based approaches were synthesised in a systematic review indicating that AI models can detect stress with acceptable overall accuracy, although sensitivity and individual variability highlight the need for cautious interpretation and human oversight (Abd-Alrazaq et al., 2024). Complementary research on embodied AI suggests that social robots may influence physiological stress regulation, producing patterns associated with calmness or positive excitement (Klęczek et al., 2024). Broader reviews and empirical studies also emphasise affective computing and emotion-recognition approaches that infer affective states in learning contexts, raising both methodological and ethical considerations (Vistorte et al., 2024). Additional strands include (a) behavioural marker approaches using smartphone or interaction data (Aalbers et al., 2023), (b) predictive analytics models for depression and suicidal ideation in university students (Meda et al., 2023), and (c) instrument development and LLM-assisted assessment (e.g., digital life balance, loneliness, and online social support) that expand the measurement toolkit but require rigorous validation (Erdemir & Atik, 2025; Gu et al., 2025).

## Theme 2 - Usage Patterns and User Characteristics

Studies depict uneven but rapidly growing uptake of AI for learning and well-being, shaped by perceived usefulness, anonymity, AI literacy, and institutional guidance. In the UAE, respondents expressed willingness to use mental-health chatbots because of accessibility and reduced stigma, but also raised concerns about privacy, trust, and accuracy (Mosleh et al., 2024). Among Chinese university students, high-frequency chatbot use was linked to emotional dependence and depressive symptoms, indicating that intensity of use can itself function as a salient behavioural marker (Zhang et al., 2025). Patterns of engagement also intersect with broader digital habits: smartphone dependency and AI-related anxiety were associated with psychological distress, sleep disruption, and concentration difficulties in student samples (Ali et al., 2025; Maraş et al., 2024). At the same time, techno-positive orientations towards AI can coexist with concerns about quality of life, inclusion, and the distribution of benefits across student groups (Lillywhite & Wolbring, 2024). Importantly, educator perspectives indicate that uptake is co-determined by institutional conditions—policies, professional development, platform integration, and governance—rather than student choice alone (Delello et al., 2025).

## Theme 3 - Perceived Benefits

Across the reviewed literature, reported benefits tended to cluster around symptom relief, enhanced self-regulation, and improved learning-related well-being. CBT-informed and generative AI chatbots were associated with reductions in depressive and anxiety symptoms and improvements in problem-solving and self-efficacy in several evaluations (Wrightson-Hester et al., 2023; Kang & Hong, 2025; Reyes-Portillo et al., 2025). Well-being platforms and AI-supported mindfulness/physical activity interventions likewise reported improvements in anxiety, sleep quality, and psychological well-being (Wang et al., 2024; Xu, 2025). In learning contexts, AI-mediated engagement and AI-enhanced feedback were linked to improved academic engagement and mental health outcomes, suggesting that reductions in cognitive burden and increases in perceived competence may be key pathways (Wang & Wang, 2024; Song, 2025; Shi et al., 2025). Evidence also points to potential gains in social connectedness and perceived support, including through LLM-assisted assessment and companionship-oriented interactions (Gu et al., 2025; Maples et al., 2024). Notably, most studies relied on self-report outcomes and short follow-up windows, highlighting the importance of cautious interpretation and longer-term evaluation.

## Theme 4 - Perceived Risks and Limitations

Risks were frequently framed in terms of technostress, anxiety, and dependence-like engagement. For example, student cohorts reported AI-related anxiety alongside only moderate levels of preparedness, indicating a training and support gap (Maraş et al., 2024). More broadly, intensive AI use was linked to digital amnesia, sleep disruption, and psychological distress, with smartphone dependency amplifying vulnerability (Ali et al., 2025). Evidence from chatbot-focused studies also suggests that frequent engagement can foster emotional reliance and correlate with higher depressive symptoms (Zhang et al., 2025). Conceptual and review-based analyses caution that rapid, unregulated AI integration may elevate stress and reduce quality of life if autonomy, boundaries, and digital balance are undermined (Klimova & Pikhart, 2025; Lițan, 2025).

In addition to behavioural risks, participants and scholars emphasised epistemic and ethical limitations. Qualitative work indicates that LLM-based offerings can generate biased or inaccurate responses and may not reliably convey empathy or clinical nuance, thereby posing potential harms if students treat outputs as authoritative (Selim et al., 2024; Connell, 2025). Privacy and surveillance concerns were recurring, especially when AI systems rely on sensitive behavioural trace data, wearable signals, or institutional analytics (Mosleh et al., 2024; Prinsloo et al., 2024). Educators also worried that over-reliance on AI could displace face-to-face help-seeking and weaken peer relationships, underscoring the need to integrate AI within stepped-care models that preserve clear escalation pathways to human support (Delello et al., 2025; Abd-Alrazaq et al., 2024).

## Theme 5 - Ethical and Policy Considerations

Ethical and policy considerations cut across all themes. Educators reported limited institutional AI policies and called for professional development and clear guidance on responsible AI use in learning and student support contexts (Delello et al., 2025). Scholarship on AI-powered educational decision support systems also highlights fiduciary responsibility, particularly for vulnerable students, and emphasises transparency, proportionality, and robust data governance (Prinsloo et al., 2024). Ethical analyses further warn against treating students as mere data points, advocating participatory design approaches and person-centred evaluation of AI systems (Skorburg et al., 2024). Finally, concerns around disability inclusion, privacy, and data security emerged as prerequisites for legitimate implementation across educational contexts (Lillywhite & Wolbring, 2024; Walan, 2025). Collectively, the evidence suggests that policy and governance cannot be treated as add-ons: they shape whether AI is experienced as supportive infrastructure for digital coping and help-seeking or as a source of surveillance and technostress.

Taken together, these themes indicate that AI-mediated cyber behaviours in higher education sit at the intersection of psychological support, learning design, and socio-technical governance. The discussion now interprets these findings in relation to the four research questions and derives implications for research and practice.

## DISCUSSION

### RQ1: AI Applications Used in Higher Education to Influence or Monitor Students' Mental Health and Well-Being

RQ1 asked how AI is being operationalised to influence student mental health in higher education. The evidence indicates two broad operational modes. First, student-facing interventions provide psychoeducational or therapeutic support through conversational agents and structured well-being platforms. Examples include CBT-informed chatbots (e.g., MYLO and Dr CareSam) and generative AI chatbots designed for reflection and mental wellness, alongside AI-supported mindfulness and physical activity interventions that embed coping resources within learning or campus life (Wrightson-Hester et al., 2023; Kang & Hong, 2025; Reyes-Portillo et al., 2025; Wang et al., 2024; Xu, 2025). Second, institution-facing assessment and monitoring applications use behavioural trace data, wearable signals, and predictive modelling to detect stress and identify risk patterns, thereby connecting AI–well-being work to learning analytics and related forms of digital phenotyping (Aalbers et al., 2023; Abd-Alrazaq et al., 2024; Meda et al., 2023; Prinsloo et al., 2024). Across both modes, AI is embedded within students' cyber behaviours: help-seeking, self-disclosure, and coping are enacted through digitally mediated interactions, while the same interactions generate data that may be used for well-being inference and decision support.

Operationalisation also varies by setting and governance arrangements. In contexts where counselling services are overstretched, chatbots are framed as scalable, low-threshold access points; in contrast, analytics-driven systems raise questions about proportionality, consent, and fiduciary responsibility. Educator perspectives underscore that implementation is mediated by institutional policy, professional development, and integration into existing platforms (Delello et al., 2025). The implication is that AI's influence on student mental health cannot be understood solely at the level of algorithmic capability; it must be analysed as a socio-technical practice embedded in institutional infrastructures and cyber behaviour ecologies.

## RQ2: Reported Effects of AI Applications on Psychological Outcomes

Evidence from the reviewed studies suggests that AI-mediated interventions can yield improvements in psychological outcomes, including reductions in anxiety and depressive symptoms, enhanced self-efficacy, and improvements in sleep quality and well-being (Wrightson-Hester et al., 2023; Kang & Hong, 2025; Reyes-Portillo et al., 2025; Wang et al., 2024). AI-enabled physical education and learning tools were also linked to increased enjoyment, engagement, and perceived well-being benefits (Xu, 2025; Song, 2025). At the same time, most evidence is based on short-term designs and self-report measures, so reported effect sizes should be interpreted cautiously and situated within the specific implementation conditions of each intervention.

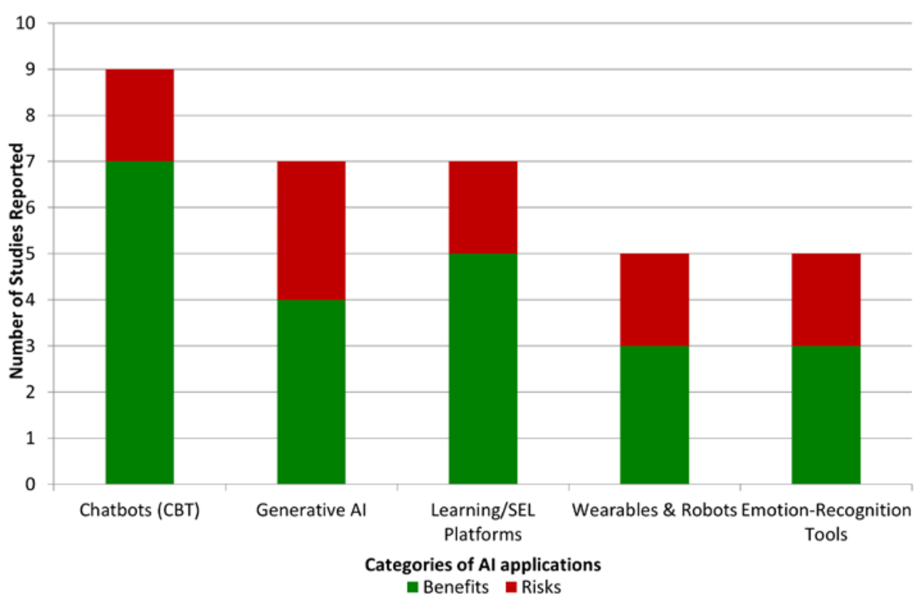
Evidence also indicates potential utility for AI-based measurement and monitoring, but with important validation and governance caveats. Wearable AI models can detect stress with acceptable overall accuracy while still showing variability across individuals and contexts (Abd-Alrazaq et al., 2024). Predictive analytics models have also been used to identify risk factors for depression and suicidal ideation among university students (Meda et al., 2023), and digital marker studies link smartphone-based patterns to stress and well-being indicators (Aalbers et al., 2023). At the instrument level, an AI-Digital Life Balance Scale extends measurement options for understanding AI-related well-being trade-offs (Erdemir & Atik, 2025), and LLM-assisted scoring has shown strong agreement for loneliness and online social support measures (Gu et al., 2025). Reviews of affective computing in educational contexts further highlight both methodological promise and risks of misclassification (Vistorte et al., 2024) (*see* Figure 3).

Importantly, beneficial effects appear contingent on baseline characteristics, AI literacy, self-regulation, and the intensity of AI engagement (*see* Table 3). For instance, frequent chatbot use may foster emotional dependence and correlate with higher depressive symptoms, suggesting that the same technology can be supportive or harmful depending on patterns of use (Zhang et al., 2025). Similarly, associations between smartphone dependency, AI-related anxiety, and psychological distress indicate that broader digital habits can amplify risk (Ali et al., 2025; Maraş et al., 2024). Review-based scholarship also cautions that unbounded AI integration may heighten technostress and reduce quality of life if boundaries and autonomy are undermined (Klimova & Pikhart, 2025; Lițan, 2025).

*Table 3. Benefits and Risks of AI Applications in Student Mental Health*

| AI Application | Benefits (n) & Examples | Risks (n) & Examples |
|---|---|---|
| **Chatbots (CBT)** | 7 – ↓ depression, ↓ anxiety, ↑ resilience, ↑ self-efficacy, ↑ social support, ↑ engagement | 2 – ↑ dependency, ↑ technostress |
| **Generative AI** | 4 – ↓ depression, ↓ anxiety, ↑ self-reflection, ↑ engagement | 3 – ↑ dependency, ↑ digital amnesia, ↓ empathy |
| **Learning/SEL Platforms** | 5 – ↓ stress, ↑ resilience, ↑ social support, ↑ well-being, ↑ academic success | 2 – ↑ technostress, ↓ human connection |
| **Wearables & Robots** | 3 – ↑ well-being, ↑ personalised recommendations, ↑ stress management | 2 – ↑ digital amnesia, ↑ somatic symptoms |
| **Emotion-Recognition Tools** | 3 – ↑ reliability of assessments, ↑ early detection, ↑ monitoring accuracy | 2 – ↑ privacy risks, ↑ bias concerns |

*Figure 3. Evidence Map of Reported Benefits and Risks of AI Applications for Student Mental Health*



## RQ3: Moderators and Mediators of AI's Impact on Student Mental Health

Individual differences in readiness and psychosocial resources emerged as plausible moderators of AI interventions. Students with higher emotional intelligence and stronger self-regulated learning capacities may be better positioned to use AI tools strategically (e.g., as bounded coping supports) rather than in dependence-like ways (Kang & Hong, 2025; Wang & Wang, 2024). AI literacy also appears important: in AI-supported learning contexts, higher AI literacy was associated with improved performance and well-being, suggesting that literacy may buffer technostress and support more reflective, agentic use (Shi et al., 2025). Conversely, baseline distress and broader digital habits may amplify vulnerability, particularly when AI engagement becomes intensive or substitutes for human support (Ali et al., 2025; Zhang et al., 2025).

Contextual factors were equally salient. Cultural norms regarding stigma, privacy, and help-seeking can shape uptake and perceived legitimacy of AI-mediated support. For example, willingness to use mental-health chatbots in the UAE was linked to the perceived anonymity and accessibility of the technology, while privacy and trust concerns constrained acceptance (Mosleh et al., 2024). In China, intensive chatbot use was associated with emotional dependence and depressive symptoms, suggesting that platform ecosystems and normative engagement patterns may shape outcomes (Zhang et al., 2025). Institutional conditions also function as moderators: educators highlighted that limited policies and professional development can heighten uncertainty and stress and lead to inconsistent adoption (Delello et al., 2025).

Critically, the evidence base rarely disaggregates effects by disability status, socio-economic background, language, or other dimensions of difference, limiting confidence about equity impacts. Yet conceptual analyses of vulnerable student digital well-being in AI-powered educational decision support systems call attention to fiduciary responsibility and the possibility that datafied systems can reproduce inequities if accessibility, consent, and accountability are not foregrounded (Prinsloo et al., 2024). Related work on technology-mediated learning indicates that students with special educational needs may experience distinctive well-being burdens, underscoring the importance of inclusive design and targeted supports (Abusneneh & Charles, 2023). Disability-focused perspectives further caution that AI's benefits may not translate into a "good life" if participation and autonomy are compromised, reinforcing the need to evaluate AI interventions across diverse student groups (Lillywhite & Wolbring, 2024).

## RQ4: Gaps, Methodological Limitations and Implications for Research and Practice

While the proliferation of AI research is encouraging, the evidence base remains methodologically heterogeneous. Much of the literature comprises proof-of-concept evaluations, cross-sectional surveys, or short-term trials, which makes it difficult to compare interventions and draw conclusions about durable effects. In addition, many studies emphasise individual-level outcomes without fully specifying the socio-technical conditions of deployment (e.g., institutional policy, governance, and platform integration). This limitation is particularly salient for monitoring and decision-support applications, where inferences about well-being depend on data quality, consent, and how institutions translate analytics into action (Prinsloo et al., 2024; Abd-Alrazaq et al., 2024). Future work would benefit from multi-site studies, longer follow-up windows, and transparent reporting of both benefits and adverse effects.

Methodological issues also extend to measurement and analytics. Many studies rely on self-report measures that may be sensitive to novelty effects or social desirability, while affective computing approaches risk misclassification when trained on limited or non-representative samples (Vistorte et al., 2024). Emerging tools—such as the AI-Digital Life Balance Scale—provide more targeted measurement of AI-related well-being trade-offs, but require further cross-cultural validation and evidence of measurement invariance across groups (Erdemir & Atik, 2025). LLM-assisted scoring of loneliness and online social support measures has also shown strong agreement with traditional scoring, yet such approaches still require careful calibration, transparency, and independent validation before being used for high-stakes decision-making (Gu et al., 2025).

From a practical perspective, institutions should avoid viewing AI as a standalone product and instead approach it as part of cyber behaviour governance for learning and well-being. Implementation challenges include interoperability with existing IT systems (e.g., learning management systems, student portals, counselling workflows), data security, privacy protection, and clear policies on data ownership, retention, and escalation to human support. Educator perspectives indicate that many settings lack coherent policies and staff development, which can increase uncertainty and stress for both educators and students (Delello et al., 2025). Ethical scholarship emphasises fiduciary responsibility and participatory design so that AI does not reduce students to "data points" and so that vulnerable learners are protected through proportionality, transparency, and accountability (Prinsloo et al., 2024; Skorburg et al., 2024). In mental health contexts, AI should complement—not replace—human-delivered support, with well-defined safeguards and escalation pathways when risk is detected or when AI outputs are uncertain (Connell, 2025).

## Overall Synthesis and Recommendations

The corpus of research reviewed here suggests that AI can both support and undermine student mental health and well-being, depending on design choices, usage patterns, and institutional governance. On one hand, AI can expand access to low-threshold coping supports and enable earlier identification of distress through validated monitoring and analytics (Wrightson-Hester et al., 2023; Abd-Alrazaq et al., 2024). On the other hand, intensive or unregulated use can amplify technostress, dependence, and privacy risks (Ali et al., 2025; Zhang et al., 2025; Prinsloo et al., 2024). Overall, the evidence points to the need for balanced, equity-oriented implementation in which AI is embedded within supportive ecosystems, bounded by clear policies and human oversight, and evaluated across diverse student groups (Delello et al., 2025; Lillywhite & Wolbring, 2024).

In light of the synthesis, recommendations are offered for stakeholders across higher education. Because AI can operate both as a supportive coping resource and as a source of technostress or surveillance, implementation should be approached as a socio-technical change process involving governance, inclusive design, and rigorous evaluation rather than as a purely technological upgrade.

## Recommendations for Higher-Education Governance and Implementation

- Establish an institutional governance framework for AI-enabled well-being tools that specifies accountability, risk assessment, audit processes, and student participation (Prinsloo et al., 2024; Skorburg et al., 2024).
- Prioritise secure integration with existing IT systems (e.g., learning management systems and student-support workflows), including data minimisation, clear retention policies, and privacy-by-design controls (Prinsloo et al., 2024).
- Provide staff development and policy guidance for educators and professional staff, addressing reported uncertainty and inconsistent institutional support (Delello et al., 2025).

## Recommendations for Student Support Services and Counselling

- Position AI chatbots as low-threshold psychoeducational and self-management supports within stepped-care pathways, with explicit escalation routes to human professionals when risk is detected or when outputs are uncertain (Connell, 2025).

- Implement safeguards for crisis scenarios, bias monitoring, and content accuracy; avoid high-stakes reliance on unvalidated outputs in clinical contexts (Selim et al., 2024; Connell, 2025).
- Monitor for dependence-like engagement and digital imbalance, particularly among students with broader vulnerability factors (Ali et al., 2025; Zhang et al., 2025; Erdemir & Atik, 2025).

### Recommendations for Teaching and Learning Design

- Apply CLT-informed design principles to reduce extraneous cognitive load and prevent AI-related overload (Song, 2025).
- Embed AI literacy, self-regulation support, and reflective use practices into curricula to mitigate technostress and support well-being (Shi et al., 2025).
- Where appropriate, integrate evidence-informed well-being interventions (e.g., mindfulness platforms and SEL approaches) into learning environments in ways that complement academic goals (Wang et al., 2024; Zong & Yang, 2025; Xu, 2025).

### Recommendations for AI Developers and Vendors

- Adopt participatory, person-centred design practices that treat students as stakeholders rather than as data points, with particular attention to vulnerable groups and accessibility (Skorburg et al., 2024; Prinsloo et al., 2024; Lillywhite & Wolbring, 2024).
- Provide transparent documentation of model behaviour, limitations, and data practices, enabling institutions to conduct informed procurement and risk assessment (Prinsloo et al., 2024).

### Recommendations for Future Research

- Conduct longitudinal and multi-site studies that evaluate sustained outcomes, adverse effects, and real-world implementation constraints across diverse student populations (Klimova & Pikhart, 2025).
- Strengthen validation of monitoring and measurement approaches, including cross-cultural testing and calibration of wearable, affective, and LLM-assisted assessments (Abd-Alrazaq et al., 2024; Vistorte et al., 2024; Gu et al., 2025).
- Prioritise subgroup analyses (e.g., disability status, language background, socio-economic factors) to test whether benefit–risk trade-offs vary across

students and to inform equity-oriented design (Abusneneh & Charles, 2023; Lillywhite & Wolbring, 2024).

Together, these recommendations aim to maximise the mental health benefits of AI while minimising technostress, inequity, and governance risks in higher education.

## CONCLUSION

The primary aim of this systematic review was to synthesise how artificial intelligence technologies affect the mental health and well-being of students in higher education. Across the reviewed literature, AI is being operationalised both as a student-facing support mechanism (e.g., conversational agents, mindfulness platforms, AI-enhanced learning tools) and as an institution-facing monitoring and analytics infrastructure (e.g., wearables, affective analytics, predictive modelling). Importantly, these deployments are enacted through and shape students' cyber behaviours: AI becomes a medium for help-seeking, self-disclosure, and coping, while also producing behavioural trace data that may be interpreted as signals of well-being.

Overall, intervention-oriented applications reported promising benefits, including reductions in anxiety and depressive symptoms, improvements in problem-solving and self-efficacy, and gains in sleep quality, engagement, and perceived well-being (Wrightson-Hester et al., 2023; Kang & Hong, 2025; Reyes-Portillo et al., 2025; Wang et al., 2024; Xu, 2025). Monitoring-oriented applications also show potential for earlier identification of distress through validated stress detection and predictive analytics (Abd-Alrazaq et al., 2024; Meda et al., 2023). At the same time, the evidence base is heterogeneous and often short-term, and reported risks are substantive: technostress, dependence-like engagement, digital fatigue, and privacy and bias concerns were recurrent across contexts (Ali et al., 2025; Zhang et al., 2025; Selim et al., 2024; Prinsloo et al., 2024; Klimova & Pikhart, 2025).

These findings have important implications for educators, counsellors, and policy makers. AI should be treated as a complement to—rather than a replacement for—human support, with clear escalation pathways when risk is detected or when AI outputs are uncertain (Connell, 2025). Institutions should develop coherent policies, invest in staff development, and ensure that AI tools integrate securely with existing IT systems and student-support workflows (Delello et al., 2025). Equity and inclusion are central: AI systems must be evaluated across diverse student groups and designed to support accessibility and participation, particularly for students with disabilities or other vulnerabilities (Abusneneh & Charles, 2023; Prinsloo et al., 2024; Lillywhite & Wolbring, 2024). Finally, participatory governance and person-centred ethics are necessary to avoid reducing students to "data points" and to ensure that AI-mediated student well-being initiatives align with institutional duties of care (Skorburg et al., 2024).

## Limitations

Several limitations qualify these conclusions. First, although the review followed PRISMA guidance, the evidence base is dominated by cross-sectional surveys and short-term evaluations, which limits inference about sustained effects and causal mechanisms. Second, heterogeneity in AI modalities, outcome measures, and institutional contexts prevented meta-analytic synthesis and makes cross-study comparison difficult. Third, many studies relied heavily on self-report measures and provided limited reporting on adverse events, data governance practices, and real-world implementation constraints, which constrains translation to practice. Fourth, and centrally for a cyber behaviour and equity lens, few studies disaggregated outcomes by key student characteristics (e.g., disability status, socio-economic background, language background), limiting confidence about differential benefit–risk trade-offs across groups. Finally, the restriction to English-language Q1 journal articles may have excluded relevant regional evidence and practice-oriented reports on institutional implementations.

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# APPENDIX A

*Table A. Detailed study characteristics for the included evidence base*

| Author(s), Year | Country | Sample Size & Population | AI Application | Study Design | Key Outcomes | Quality / Limitations |
|---|---|---|---|---|---|---|
| **He et al., 2022** | China | 185 university students | "XiaoE" CBT chatbot | RCT (3 arms) | ↓ depression and anxiety at 1 month vs. controls | Short follow-up; single context |
| **Liu et al., 2022** | China | 120 undergraduates | CBT chatbot (self-help) | RCT | Greater ↓ in depression/anxiety vs. bibliotherapy | Unblinded; small sample |
| **Wrightson-Hester et al., 2023** | UK | 50 adolescents & students | "MYLO" chatbot | Case series | ↓ distress, ↑ goal conflict resolution | No control group |
| **Kang & Hong, 2025** | Korea | 98 students | "Dr CareSam" chatbot (ChatGPT-4) | Mixed methods trial | High satisfaction; ↑ therapeutic alliance | Cultural/ contextual limits |
| **Reyes-Portillo et al., 2025** | USA | 210 students (distressed sample) | "Wayhaven" LLM chatbot | Open trial | ↓ depression & anxiety, ↑ agency | Uncontrolled; attrition |
| **Dechant et al., 2025** | USA | 180 youth/ students | "Future Me" chatbot | Exploratory UX | ↑ goal-setting, but ↓ engagement over time | Declining use; low empathy |
| **Maples et al., 2024** | Global survey | 3,000+ chatbot users | "Replika" | Cross-sectional survey | ↑ social support; 3% suicide prevention | Self-report; selection bias |
| **Chen et al., 2024** | China | Bibliometric (n=248 articles) | N/A | Bibliometric analysis | Rapid rise in AI–mental health research | Descriptive only |
| **Gu et al., 2025** | China | 200+ students | ChatGPT-generated scales | Cross-sectional | Strong correlations with validated tools | Needs replication |
| **Ali et al., 2025** | Egypt | 500 nursing students | Smartphone AI | Cross-sectional survey | ↑ smartphone dependence ↔ digital amnesia, somatic symptoms | Non-causal |
| **Song, 2025** | Europe (music education) | NR; music education students | AI-powered learning tools | Quantitative survey | ↓ cognitive load; ↑ well-being; ↑ academic success | Context-specific; generalisability uncertain |
| **Zong & Yang, 2025** | China (EFL context) | NR; EFL students | AI-enhanced SEL framework | Quantitative study | ↑ engagement; ↑ emotional well-being | Indirect well-being measures; limited controls |
| **Wu, 2024** | NR (Asia) | NR; foreign-language learners | AI-supported foreign language environment | Quantitative study | ↑ adaptability; anxiety/ loneliness/depression still impaired learning | Non-causal; outcome measures limited |
| **Shi, 2025** | NR | NR; HE students | Generative AI–supported writing | Cross-sectional modelling | AI literacy & self-regulation → ↑ writing performance → ↑ digital well-being | Cross-sectional; cannot infer causality |
| **Wang & Wang, 2024** | China | 2,423 undergraduates | General AI use & intention | Cross-sectional SEM | AI use/intention mediated engagement ↔ mental health | Observational; subjective norms NS |
| **Mosleh et al., 2024** | UAE | NR; undergraduates | Chatbot use (education) | Cross-sectional survey | 83.6% used chatbots; ↑ emotional intelligence ↔ chatbot use | Self-report; context-specific |

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*Table A. Continued*

| Author(s), Year | Country | Sample Size & Population | AI Application | Study Design | Key Outcomes | Quality / Limitations |
|---|---|---|---|---|---|---|
| **Ahmed et al., 2025** | NR | NR; students | AI-driven wearables + LLM insights | Preliminary study | ↑ personalised recommendations; unique emotion insights | Pilot; small sample; external validity limited |
| **Abd-Alrazaq et al., 2024** | Multi-country (19 studies) | HE students | Wearable-AI stress detection | Systematic review/meta-analysis | Accuracy 0.856; sensitivity 0.755; modest performance | Recommend pairing with clinical assessment |
| **Klęczek et al., 2024** | NR | NR; adult/ student participants | Social robot vs. (breathing exercises) | Experimental (physiological) | Robot users showed ↑ calm/ positive excitation states; self-reports similar | Small sample; HE specificity unclear |
| **Aalbers et al., 2023** | NR | College students | Smartphone ML predictors of stress | Idiographic ML analysis | Social media & sleep proxies ↔ stress; modest correlations | High variability; generalisability limited |
| **Ali et al., 2025** | Egypt | NR; nursing students | Smartphone dependency, AI-linked use | Cross-sectional survey | ↑ smartphone dependence ↔ ↑ digital amnesia & somatic symptoms | Non-causal; confounding likely |
| **Erdemir & Atik, 2025** | Turkey | 3 university samples | AI–Digital Life Balance Scale (AI-DLBS) | Psychometric validation | 6-factor structure; α=0.68–0.87; ↑ test–retest reliability | Needs cross-cultural validation |
| **Gu et al., 2025** | China | 200+ undergraduates | ChatGPT-generated loneliness/social support scales | Cross-sectional comparison | ICC = 0.81 (loneliness), 0.95 (social support) vs. validated tools | Single-country; replication needed |
| **Meda et al., 2023** | Italy | NR; undergraduates | ML predictors of severe depression/ ideation | Predictive modelling (cross-sectional) | Identified ML predictors of depression & suicidal ideation | Cross-sectional; not intervention |
| **Shahzad et al., 2024** | NR | NR; HE students | AI + social media in smart learning | Cross-sectional perceptions | Students perceived ↑ academic performance & ↑ well-being | Self-report; causal inference limited |
| **ur Rehman et al., 2024** | NR | NR; HE students | ChatGPT usage | Cross-sectional survey | ↑ life satisfaction; moderated by subjective health | Self-report; common-method bias risk |
| **Maraş et al., 2024** | NR | Nursing students & candidates | | Cross-sectional survey | Moderate AI anxiety; ↑ training need | Not exclusively HE; domain-specific |
| **Chen et al., 2024** | China | 248 articles (bibliometric) | N/A (bibliometric mapping) | Bibliometric analysis | ↑ research outputs post-2022; focus on AI & anxiety | Contextual only; not empirical |
| **Vistorte et al., 2024** | NR | 41 studies (systematic review) | AI for emotion assessment | Systematic review | Synthesised 4 themes (recognition, integration, assistive tech, ethics) | Review only; not HE-focused |
| **DeLello et al., 2025** | USA | NR; college educators (not students) | AI in classroom (policy/ethics) | Cross-sectional survey | 92% awareness; ⅓ lacked institutional AI policy; ↑ stress reported | Educator-focused; not student sample |
| **Klimova & Pikhart, 2025** | Europe | N/A | AI in higher education | Mini-review | Highlighted potential benefits; warned of ↑ technostress, privacy risks | Narrative; not primary data |

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*Table A. Continued*

| Author(s), Year | Country | Sample Size & Population | AI Application | Study Design | Key Outcomes | Quality / Limitations |
|---|---|---|---|---|---|---|
| **Roberts & Wolbring, 2024** | NR | Conceptual (philosophical) | Ethical critique of AI in HE | Theoretical essay | Warned AI → treating students as data points; ↓ participatory ethics | No empirical data |
| **Ramírez & Kness, 2025** | NR | Consulting psychologists' views | AI in interpersonal contexts | Conceptual commentary | Warned AI may distort communication & relationships | Opinion-based |
| **Prinsloo et al., 2024** | South Africa/ UK | NR | AI-EDSS systems in HE | Conceptual perspective | Advocated ↑ fiduciary responsibility; ↑ digital well-being | Not empirical; contextual |
| **Xie et al., 2025** | Ireland (medical students) | NR; planned sample | AI for contextual well-being | Protocol (mixed-methods) | Aimed to explore AI, autonomy, authenticity | Protocol only; no results |
| **Kostenius et al., 2024** | Scandinavia | Youth incl. students | Chatbot for mental well-being | Qualitative interviews | Reported ↑ acceptance; chatbot useful in remote regions | Not HE-exclusive |
| **Walan, 2025** | Sweden | Primary school pupils | AI perceptions (general) | Cross-sectional | Pupils voiced ↑ privacy/job loss concerns | Non-HE; young learners |
| **Connell, 2025** | NR | NR | Organisational consulting & AI | Perspective article | Warned of unethical AI harms | Commentary; no student sample |

**Key:** ↑ = Increase/Improvement; ↓ = Decrease/Reduction; ↔ = Association/Link; NS = Not significant; NR = Not reported